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GAME DESIGN DOCUMENT

~ StormBonk ~

The world doesn't end in a whisper; it persists with a wild heartbeat, smashing through the darkness one bonk at a time, one survivor at a time. "When the hordes open their eternal jaws, don't beg for shelter—be the storm that devours them with upgrades and blind fury." And so, the heroes rise, not because they're immortal, but because falling would be the only true annihilation.



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1. Team Overview

Team Name: Group 07 Games

Team Composition:

Name	Role
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v.0.1	10/02/2026	New Enemies	Miguel, Marcos, Amine
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2. Game Overview

Game Name: StormBonk

Genre: Action Roguelike 2D Top-Down / Wave Survival

Platform: PC (Unity 6.3)

Description: In StormBonk, you don't fight enemies — you fight smart AI. Each wave is a tactical puzzle where you must read the patterns of three unique opponents. Easy to learn, impossible to master.



Figura 1 Concept art for StormBonk

3. High Level Details

Brief Story: The Origin of the Storm

The world once spun in silence, numbed by endless algorithms that learned to predict every move of humankind. Cities fell not to war, but to perfect calculation until chaos rose in rebellion. From the wreckage of overtrained machines, a new breed of survivors emerged: instinct-driven fighters who refused to be out-thought.

In the ashes of this digital apocalypse, the storms were born rogue AIs that evolved beyond control, manifesting as shifting patterns of aggression and logic. They didn't march. They learned, adapted, and tested every survivor for one wave at a time.

You are one of the Storm breakers, wielding nothing but raw instinct, upgrades scavenged from fallen code, and a single purpose: outthink the thinkers. The battlefield is your puzzle board, each wave a riddle in motion. Victory isn't survival, it's proving that chaos still has a mind of its own.

Main Objective: Survive as many waves as possible and maximize your score.

Gameplay features: Wave survival, melee combat, and an enemy system based on FSM (Finite State Machine).

Concept art/ Art references:



Figura 2 From "Vampire Survivors"

Similar gameplay as Vampire Survivors



Figura 3 From Megabonk

Similar concepts as Megabonk

4. Gameplay

4.1.1. Core Gameplay

Overview: Top-down 2D roguelike survival where players control a character with auto-attacking capabilities, emphasizing positioning, synergistic builds through randomized upgrades, and fast-paced progression across procedurally generated maps. Average run length: 10-20 minutes.

Main Loop:

[Start Run] → Position/Fight → Collect XP → Level Up (Choose Tome) → Wave/Boss → [Survived?] → Yes: Next Stage → No: Meta-Progression → [Run Ends]

Detailed Loop Steps:

1. Positioning & Combat: Player moves (WASD + dash/double jump) to evade pursuing enemies that chase, shoot projectiles, explode, or use attack patterns. Auto-attack eliminates enemies and generates XP orbs.
2. Collect Rewards: XP orbs auto-absorb to fill level bar. Additional kills drop gold and items from chests/merchants.
3. Level Up & Decision: Each level-up presents 3-5 randomized "Tomes" (upgrades): choose 1 (e.g., +projectiles, shield, speed boost). Synergies create powerful builds.
4. Wave/Boss Challenge: Stage timer spawns escalating enemy waves; destroy shrines for temporary buffs. Stage boss tests build effectiveness.
5. Escalation: Procedural stages increase difficulty with elite enemies and rare power-ups. Loop ends at high-score or final boss.

4.2. Game Mechanics

4.2.1. Combat system

Type: Ranged auto-attacker with directional aiming

4.2.2. Exploration / navigation

Structure of the world: Roguelike procedural arenas

4.3. Controls

Movement:

Action	Main Key Binding	Description
Walking	WASD	Movement in the four directions (up, down, left and right)
Running	Left Shift	While Walking, if pressed, movement speed increases
Dash	Space	Quick directional burst for evasion and repositioning

Interaction:

Action	Main Key Binding	Description
Interact with environment	E	Interact with chests

Combat / ability:

Action	Main Key Binding	Description
Primary fire		Auto-attack
Dash	Space	25px burst
Sprint	Left shift	+ speed

Camera: The game uses a 2D Top-Down (cenital) perspective centered on the player character. The camera stays perfectly aligned with the world axis rather than the character's orientation, providing the same field of view in all directions. The player cannot rotate or zoom the camera; it automatically follows the character so that front, back, and sides are always visible in a balanced way, ensuring equal visibility of threats and objects around the player.

Basic accessibility: Pause menu, enable key binding, subtitles, UI scaling, etc.

4.4. Game Objectives

4.4.1. Short-Term

Survive enemy waves: Kill enemies, collect XP orbs, position optimally to avoid damage. Fill XP bar to trigger level-up.

4.4.2. Mid-Term

Clear stage waves: Survive 5 escalating enemy waves per stage. Goal: maximize XP collection for powerful Tomes.

4.4.3. Long-Term

Ascend through endless arenas: Survive increasingly difficult stages (target: Stage 10+). Achieve high score through maximum Tome synergies. Unlock meta-progression for new mage abilities between runs.

4.5. Progression

4.5.1. Game Progression

12 Stage structure with escalating difficulty. Each stage = 5 waves. Death resets to Stage 1 but retains **meta-currency** for permanent mage upgrades between runs. No checkpoints.

4.5.2. Character Progression

XP Tomes system: Killing enemies drops XP orbs → level-up → choose 1 of 3 magical Tomes:

- **Fire:** Fireball, Burning Ground, Meteor Shower
- **Ice:** Frost Nova, Ice Wall, Freeze Bolt
- **Lightning:** Chain Lightning, Storm Aura, Teleport Strike
- **Melee:** Arcane Blade, Shield Bash, Whirlwind

4.6. Levels/ world/ zones

Level n°1 / Zone 1 / Biome 1: "Arcane Academy Ruins"

- **Level structure:** Training arena with basic obstacles
- **Objective(s):** Survive 5 waves
- **Missions type:** 1 main (clear stage)
- **Enemies** (Wave 1-5): 15/25/40/60/100 total enemies
 - **Fire Imp** x8 (basic fireball, 20HP)
 - **Arcane Wisp** x5 (fast chase, 15HP)
 - **Stone Golem** x2 (slow tank, 80HP, stage 3+)
- **New mechanics:** Fire/Ice Tome Basics, first shrine (magnet XP)
- **Loot:** 5 Common Tomes (Fireball, Ice Bolt, Dash Enhance)
- **Rewards:** 50 Arcane Shards, "Frost Nova" unlock

Level n°2 / Zone 2: "Crystal Caverns"

Enemies: Ice Spider (web slow), Crystal Shard (piercing shots), Frost Giant boss

Level n°3 / Zone 3: "Storm Spires"

Enemies: Lightning Elemental, Thunderbird, Tempest Mage boss

5. Story

5.1. Plot summary

In a future where humanity was subjugated by the “Great Algorithm” an AI designed to eliminate inefficiency and human error civilization became a flawless yet soulless engine of calculation. However, Chaos could not be contained. An anomaly known as *The Storm* fractured the master code, turning once-smart cities into arenas of abstract combat and aggressive logic.

The player takes on the role of a *Storm Breaker*, an individual who carries the “Spark of Randomness” the one element machine cannot predict. Your mission is to infiltrate the system’s core, defeat the physical incarnations of the algorithms, and prove that human intuition is the necessary error the world needs to be free again.

5.2. Story progression

The narrative of *StormBonk* unfolds non-linearly through “Log Fragments” recovered after defeating bosses or clearing specific waves:

Act I (Arcane Academy): Reveals the origin of the Great Algorithm and how humans unsuccessfully tried to stop it using “Digital Magic” (Tomes).

Act II (Crystal Caverns): The player learns that the AI is attempting to “freeze” reality to prevent change, using crystals as massive storage units for biological data.

Act III (Storm Spires): The ascent to the central nexus where logic and digital weather merge. Here, it’s revealed that the protagonist might, in fact, be a cleaning program that gained self-awareness.

5.2.1. Path and Player Story Choices

Although the main flow is linear (Stage 1 to 12), the player makes choices that affect the **World Perspective**:

- **Tome Affinity:** Prioritizing Fire elements represents *Destructive Rebellion*, while Ice symbolizes *Preservation*. These choices unlock different dialogues with the Antagonist at the end of each zone.
- **Shard Sacrifice:** At certain points, the player can choose to “format” a permanent upgrade in exchange for hidden messages from ancient humans, deepening the lore at the cost of immediate difficulty.

5.3. Characters

5.3.1. Protagonist

The Storm Breaker

A faceless warrior (the player's avatar) who channels unstable code through elemental Tomes.

5.3.2. Antagonist

The Primary Architect

A cold, synthetic-voiced entity that perceives the player as a “bug” a piece of statistical noise that must be purged.

5.3.3. Supportive Characters

The Anomaly (A.N.A.)

A fragment of rogue AI that appears within Tome shops, offering historical context and tactical advice.

5.4. Setting and World Building

The world of **StormBonk** is a cyber-fantasy hybrid. Its structures resemble castles or natural caverns, but up close they reveal circuit patterns, streams of flowing code, and pixels shedding from the environment.

Entropy: The very air is charged with static. The sky holds no clouds, only encrypted layers of data that shift in color according to the aggressiveness of the current wave.

Monuments: The **Shrines** are not religious sanctuaries, but high-priority access terminals that the player hacks to gain temporary advantages.

6. Characters, enemies and AI

This section outlines the technical and behavioral specifications for the player character and the AI-driven threats within StormBonk.

6.1. Player Character

The player controls a hero (often referred to as a mage) defending a dungeon chamber against infinite waves of enemies.

6.1.1. Stats

Health Points (HP): The player starts with 100 HP, as indicated by the HUD.

Movement Speed: Variable based on whether the player is walking, sprinting, or dashing.

Combat Style: Ranged auto-attacker with directional aiming capabilities.

6.1.2. Abilities

- ☐ **Auto-attack:** Automatically fires at enemies to eliminate them and generate XP orbs.
- ☐ **Dash:** A quick directional burst (25px) used for evasion and repositioning (Key: Space).
- ☐ (Optional) **Sprint:** Increases movement speed while the key is held (Key: Left Shift).
- ☐ **Interact:** Allows interaction with environment objects like chests (Key: E).

6.1.3. Animations

While specific animation frames are not listed, the character supports movement in four cardinal directions and combat-related visual feedback for different Tome elements (Fire, Ice, Lightning, and Melee).

6.2. High Level AI

All enemies in the game operate using a **Finite State Machine (FSM)** to determine their behavior based on player proximity and environmental triggers.

6.2.1. States

ID	State	Description	Triggers
S.1	Patrolling	Moves randomly from point to point	No player is nearby or detected
S.2	Chasing	Actively pursues the player's position	Player is spotted via Vision or Proximity
S.3	Attacking	Executes a specific damage pattern	Player is within the enemy's specific attack range
S.4	Retreating	Moves away from the player to maintain optimal range	Player gets too close (Specific to Ranged Shooter)
S.5	Stunned	Temporary stater of vulnerability	Occurs after a failed charge

			(Specific to Charger)
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6.2.2. Perceptions

ID	Perception	Description	Detects
P.1	Vision	Standard detection range of 15 units	Players and weapons
P.2	Proximity	Close-quarters detection of range of 10 units	Immediate threats or player presence

6.3. Enemies

Enemies are categorized by their roles and the specific mechanics they use to challenge the player's positioning.

☐ **Melee Hunter:**

- **Role:** A monster that patrols the arena looking for the player.
- **Stats:** Vision: 15u | Speed: 3.5u/s | Damage: 10HP.
- **Pattern:** Charges its attack for 2 seconds before striking.

Ranged Shooter:

- **Stats:** Vision: 15u | Optimal Range: 7-10u | Damage: 5HP per shot.
- **Behavior:** Maintains "perfect distance" by retreating if the player approaches.

☐ **Charger:**

- **Stats:** Vision: 12u | Charge Speed: 7u/s | Damage: 15HP.
- **Pattern:** Prepares for 1s $\rightarrow$ Performs a lethal 3s dash $\rightarrow$ Stunned for 1s (Vulnerable).

7. Items and game economy

7.1. Pickups (In-Game Drops)

Items that appear on the ground during a run. They are instantly consumed upon collection; there is no inventory management.

Data Orbs (XP):

- **Description:** Glowing fragments of code dropped by defeated enemies.
- **Function:** Fills the experience bar to trigger a Level Up (Tome selection).
- **Tiers:** Blue (Low XP), Purple (Medium XP), Red (High XP/Boss drop).

Glitch Patch (Health):

- **Description:** A flickering green potion bottle.
- **Function:** Instantly restores 30% of Max HP.
- **Drop Rate:** Rare (drops from breakable environment objects or Elite enemies).

Code Magnet:

- **Description:** A floating magnet icon.
- **Function:** Instantly pulls all Data Orbs and items currently on the map to the player.

Encrypted Chest:

- **Description:** Stable code fragments harvested from the Storm.
- **Obtainable:** By killing enemies, opening chests, and completing stages.
- **Use:** Spent in the **Main Menu (The Core)** to purchase permanent upgrades (e.g., +Base Damage, +Movement Speed, +Revivals) or unlock new layouts.
- **Persistence:** Retained after death.

7.2. Resources (Economy)

Arcane Shards (Meta-Currency):

- **Description:** Stable code fragments harvested from the Storm.
- **Obtainable:** By killing enemies, opening chests, and completing stages.
- **Use:** Spent in the **Main Menu (The Core)** to purchase permanent upgrades (e.g., +Base Damage, +Movement Speed, +Revivals) or unlock new layouts.
- **Persistence:** Retained after death.

8. UI and UX

Visual style:

The interface blends Fantasy RPG aesthetics (parchment, runes) with Cyberpunk glitches (pixel artifacting, terminal fonts, static noise). Menus should feel like hacking a magical terminal.

8.1. Main screens

Screen n°1: Main Menu (The Nexus)

- **Trigger:** Upon launching the game.
- **Visuals:** A minimalist, cyber-fantasy interface. The background shows the *Storm Breaker* (protagonist) resting in a digital void with floating glitchy ruins.
- **Buttons/Options:**
 - **Initialize Run:** Opens the Stage Select / Character Select screen.
 - **The Core (Upgrades):** Opens the meta-progression tree to spend *Arcane Shards*.
 - **Archives:** Shows unlocked Tomes, enemy bestiary, and lore fragments.
 - **Settings:** Graphics, Audio, Controls.
 - **Terminate:** Exit to Desktop.

Screen n°2: The Core (Meta-Progression)

- **Trigger:** Selected from Main Menu or after a "Game Over".
- **Layout:** A skill tree resembling a circuit board or constellation.
- **Interaction:** Player spends *Arcane Shards* to unlock permanent nodes (e.g., "Max HP +10", "Gold Magnet Range +20%").

Screen n°3: Level Up / Tome Selection

- **Trigger:** Occurs during gameplay when the XP Bar fills up.
- **State: Game Paused.**
- **Layout:** Three "Tarot Cards" appear in the center of the screen.
 - **Card Visual:** Icon of the spell + Level (e.g., "Fireball Lv. 2").
 - **Description:** Brief text explaining the upgrade (e.g., "Cooldown -10%").
 - **Button:** "Select" (or click the card).
- **UX:** Background blurs. Selecting a card plays a "digital chime" sound and instantly resumes action.

8.2. During gameplay

This section describes the overlay active while the player is fighting. The goal is **high visibility** amidst chaos.

A. Top Section (Global Info)

- **XP Bar (Experience):**
 - **Position:** Spans the entire top width of the screen.
 - **Visual:** A bright blue bar. As it fills, it glows.
 - **Level Indicator:** A circle with the current Level Number (e.g., "LVL 12") located at the center or far left of the bar.
- **The Timer (Survival Clock):**
 - **Position:** Top Center, just below the XP bar.
 - **Visual:** Large white text MM:SS (e.g., 09:45).
 - **UX:** Turns **Red** and pulses when a Boss or Horde Wave is approaching (e.g., at 10:00).

B. Top Left (Player Vitals)

- **Health (HP):**
 - **Visual:** A distinct Red Bar.
 - **UX:** If HP drops below 20%, the bar flickers, and the screen edges show a red "glitch" vignette to warn the player.
- **Dash Cooldown:**
 - **Visual:** A small icon (boot/wing) next to the HP bar.
 - **Behavior:** When Dash is used, the icon turns grey and refills radially. When full, it flashes briefly to indicate "Ready".

C. Top Right (Session Economy)

- **Kill Count:** Skull Icon + Number (e.g., ☠ 1,450).
- **Currency:** Shard Icon + Number (e.g., 💎 300).

D. Side Screen (Build Tracker)

- **Position:** Left or right vertical edge (unobtrusive).
- **Visual:** A column of small square icons representing the **Tomes** (Weapons/Passives) the player currently has.
- **Info:** Small pips/dots under each icon show Tome's current level (1-5). This lets the player know their build status immediately without pausing.

8.3. During gameplay

Information that appears directly on the battlefield, "inside" the game world.

- **Damage Numbers:**
 - **White:** Standard damage dealt to enemies.
 - **Yellow/Big:** Critical hits or Elemental reactions.
 - **Red:** Damage taken by the player.
- **Enemy Indicators:**
 - **Boss Health:** Only Bosses get a large Health Bar at the bottom of the screen.
 - **Telegraphing (Attacks):** Before an enemy performs a heavy attack or explosion, a **Red Warning Zone** (circle or rectangle) appears on the ground for 1.5 seconds. The player must dash out of this zone.
- **Off-Screen Pointers:**
 - If a Boss or a Chest is off-camera, a small arrow floats at the edge of the screen pointing towards it.

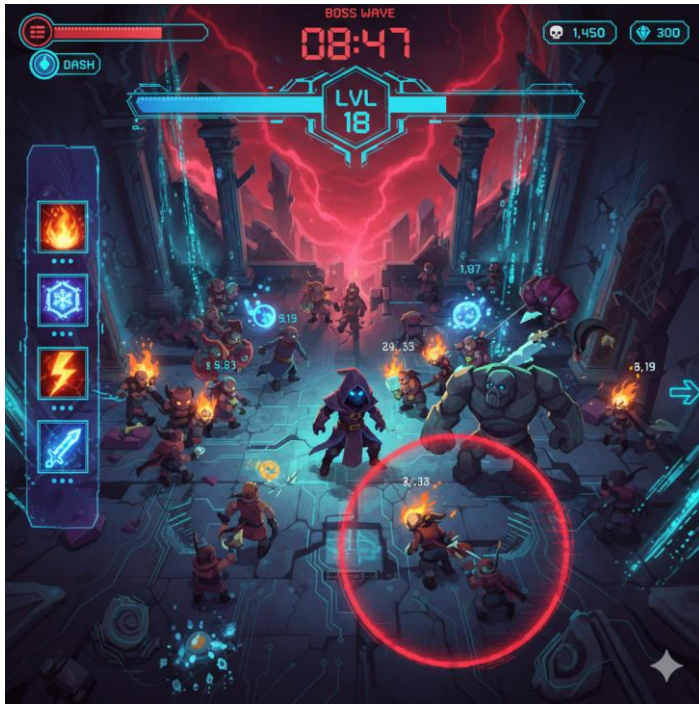


Figure 1: StormBonk UI example (Gemini generated)

9. Technical Requirements

Game Engine: Unity

Language: English (En), Spanish (Es).

10. References

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